

B3.2 Coordination and control – the endocrine system

Summary

Hormones are chemical messengers. In animals, hormones are transported around the body in the blood and affect target tissues and organs. Hormones have a variety of roles in the human body, including controlling reproduction. Plant hormones are chemicals that regulate plant growth and development. They can be used in agriculture to control the rate of growth.

Underlying knowledge and understanding

Learner should be aware of a number of hormones including adrenaline and the male and female sex hormones.

Common misconceptions

With regards to the menstrual cycle, research has shown that learners have problems relating the time of conception to the condition of the lining of the uterus.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM3.2i	extract and interpret data from graphs, charts and tables	M2c
BM3.2ii	translate information between numerical and graphical forms	M4a

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B3.2a describe the principles of hormonal coordination and control by the human endocrine system	use of chemical messengers, transport in blood, endocrine glands and receptors			
B3.2b explain the roles of thyroxine and adrenaline in the body	thyroxine as an example of a negative feedback system			

B3.2c	describe the role of hormones in human reproduction including the control of the menstrual cycle	oestrogen, progesterone, FSH and testosterone		WS1.3b, WS1.3e	
B3.2d	explain the interactions of FSH, LH, oestrogen and progesterone in the control of the menstrual cycle		M2c, M4a, M2g		Analysis of relative hormones levels from raw data and graphically.
B3.2e	explain the use of hormones in contraception and evaluate hormonal and non-hormonal methods of contraception	relative effectiveness of the different forms of contraception	M2c, M4a	WS1.1d, WS1.1e, WS1.1f	Discussion into the various methods of contraception and their effective/ethical use.
B3.2f	explain the use of hormones in modern reproductive technologies to treat infertility			WS1.1d, WS1.1e, WS1.1f, WS1.1h	Research into <i>Xenopus laevis</i> pregnancy testing to detect hCG by the stimulation of oogenesis. Research into hormonal treatments for infertility.
B3.2g ☑	explain how plant hormones are important in the control and coordination of plant growth and development, with reference to the role of auxins in phototropisms and gravitropisms	unequal distribution of auxin		WS2a, WS2b, WS2c, WS2d	Investigation of the effects of phototropism using seedlings. (PAG B6)
B3.2h ☑	describe some of the variety of effects of plant hormones, relating to auxins, gibberellins and ethene	controlling growth, controlling germination, fruit ripening, flower opening and shedding of leaves		WS2a, WS2b, WS2c, WS2d	Investigation/research into the question 'does one bad banana spoil the fruit bowl?' (PAG B2, PAG B6)
B3.2i ☑	describe some of the different ways in which people use plant hormones to control plant growth	selective herbicides, root cuttings, seedless fruit (parthenocarpic fruit development), altering dormancy			